Mavangwa Andzani Singh

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PROFESSIONAL SUMMARY

Software Developer and IT Tutor currently pursuing a Diploma in Information Technology (Software Development) at Vaal University of Technology. Passionate about building innovative digital solutions, exploring Artificial Intelligence and machine learning and developing modern, user-friendly web and desktop applications. Interested in both frontend and backend development, with a strong focus on continuous learning, problem-solving and using technology to address real-world challenges.

EXPERIENCE

Undergraduate Personal Tutor 08/2025 – Present
Vaal University of Technology Vanderbijlpark, Gauteng

Information System Information Systems & System Design

- Tutor students in Systems Analysis and Design.
- Assist students with system design, Data Flow Diagrams (DFDs) and Entity Relationship Diagrams (ERDs).
- Guide students in analyzing system requirements and designing information systems.
- Mentor students on academic projects, including Hospital Management Systems.

Software Development & Databases

- Tutor students in VB.NET, SQL and PL/SQL.
- Assist with SQL queries, stored procedures and database development.
- Support students with debugging, programming concepts and application development.
- Guide students through practical exercises, assignments and semester projects.

EDUCATION

Diploma in Information Technology (Software Development) 02/2024 – Present
Vaal University of Technology Vanderbijlpark, Gauteng

National Senior Certificate (Matric) 11/2023
Kheto Nxumayo Agricultural High School Giyani, Limpopo

TECHNICAL SKILLS

Languages
Java, Python, SQL, PHP, SQL (Postgres), JavaScript, HTML, CSS, VB.Net

Developer Tools
Git, GitHub, VS Code, Visual Studio, Net Beans, Note Pad ++

Frameworks & Technologies
React, Fast API, Rest API

Databases
MySQL, PLSQL, Firebase

CERTIFICATES

IBM Skills-Build - Data Science Tools
March 2024 - March 2024

FNB App Academy - Full- Stack Development
May 2025-June 2025

IBM Skills-Build- Cybersecurity Fundamentals
June 2024-June 2024

FNB App Academy - Full- Stack App Development
July 2026 - Present

PROFILE & SKILLS

- Home Language: Xitsonga
- Other Language(s): English(fluent)
- Communication
- Problem Solving
- Teamwork
- Time Management
- Creativity and Innovation
- Adaptability
- Willingness to Learn
- Leadership

PROJECTS

Residence Management System 09/2026 – 10/2026

HTML, CSS, JavaScript, PHP, JSON

- Developed and deployed a full-stack website for managing student residence applications and accommodation information.
- Implemented student registration, authentication, and application management functionality.
- Used PHP and JSON to process applications and manage residence data.
- Created a responsive interface for desktop and mobile devices

AgriScan AI – Plant Disease Detection System 08/2026 – 10/2026

Python, TensorFlow, CNN, Numpy, Deep Learning

- Developed a Python-based AI system to detect plant diseases from leaf images.
- Built and trained a Convolutional Neural Network (CNN) model from scratch for image classification.
- Prepared and processed plant image datasets to train and evaluate the model.
- Evaluated model performance using classification metrics and visualizations.
- Applied deep learning techniques to support early plant disease identification.

Residence Maintenance Reporting System 05/2026 – 06/2026

HTML, CSS, JavaScript, PHP, JSON

- Developed and deployed a full-stack website for reporting, managing, and tracking student residence maintenance issues.
- Implemented role-based access for students, maintenance staff, and administrators.
- Used PHP for backend processing and JSON for data storage and management.
- Designed a responsive interface for desktop and mobile devices.

Java Library Management System 05/2026 – 06/2026

Java 21, Swing, FlatLaf, NetBeans

- Developed a desktop application to automate library operations and reduce manual record-keeping.
- Implemented book management, including adding, editing, deleting, searching, and tracking availability.
- Built user authentication, registration, and role-based access for administrators and users.
- Implemented book borrowing, returns, and transaction history using in-memory data storage.

Property Maintenance System 04/2026 – 06/2026

HTML, CSS, JavaScript, PHP, JSON

- Developed and deployed a full-stack web application for managing property-related information and operations.
- Built responsive user interfaces and implemented backend functionality using PHP.
- Used JSON for storing and managing application data.
- Designed user-friendly interfaces to simplify property management tasks

REFERENCES

Ms Rikhotso TO, *Administrator of ICT,* VUT (Vanderbijlpark CAMPUS)
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